ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET first paper. Section A (35): multiple choice, true or false, matching and choosing the correct word · Section B (65): the parts and safe use of a 3D printer, a Raspberry Pi conveyor program (gpiozero) to study, current, power, energy and parallel resistance, and the laws of robotics in a mine-rescue scenario. Formula sheet at the end.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B, and a formula sheet.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which sensor tells a robot that it has bumped into something?

A Ultrasonic sensor
B PIR sensor
C Touch (bumper) switch
D LDR

1.2 Which type of robot is fixed to the floor and used on assembly lines?

A Industrial robot arm
B Planet rover
C Drone
D Humanoid

1.3 The part of a 3D printer that heats and melts the filament is the …

A print bed.
B hot end (nozzle).
C stepper motor.
D frame.

1.4 A Raspberry Pi is normally powered by …

A a 230 V plug connected directly.
B a 9 V battery clip.
C a 12 V car socket.
D a 5 V USB power supply.

1.5 Which component changes electrical energy into movement?

A Motor
B LED
C Resistor
D Capacitor

1.6 A slicer program produces …

A an STL model.
B a photograph.
C G-code for the printer.
D Python code.

1.7 Which Python statement pauses a program for one second (after importing time)?

A wait(1)
B time.sleep(1)
C pause()
D delay(1000)

1.8 Which law of robotics says that a robot must protect its own existence, as long as this does not conflict with the other laws?

A First law
B Second law
C Fourth law
D Third law

1.9 Electric current is measured in …

A amperes.
B volts.
C ohms.
D watts.

1.10 Which ONE of the following is an actuator?

A Thermistor
B Servo motor
C LDR
D Microphone

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 Robots never need maintenance.

2.2 A PIR sensor detects the infrared radiation given off by warm bodies.

2.3 A Raspberry Pi GPIO pin can be damaged if a 5 V signal is connected to it.

2.4 PLA warps more easily than ABS.

2.5 The power rails of a breadboard run along its long edges.

2.6 A buzzer is an input device.

2.7 Voltage is the "push" that makes current flow in a circuit.

2.8 Robots can do work in places that are dangerous for people.

2.9 Adding a resistor to a circuit increases the current.

2.10 Python uses indentation to show which lines belong inside a loop.

(10)

[10]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Robot that moves around on wheels or tracks	A	diode
3.2	Small computer on one chip, used to control devices	B	program
3.3	Device that measures how far a shaft has turned	C	mobile robot
3.4	Set of instructions that a computer follows	D	heated bed
3.5	Unit of electrical power	E	watt
3.6	Warm platform that helps stop a 3D print from warping	F	humanoid
3.7	Wire used to connect points on a breadboard	G	microcontroller
3.8	Robot designed to look and move like a person	H	teleoperation
3.9	Component that lets current flow in one direction only	I	jumper wire
3.10	Controlling a robot by a person from a distance	J	encoder

(10)

[10]

QUESTION 4

Complete the following sentences by choosing the correct word in brackets. Write only the word next to the question number (4.1–4.5) in the ANSWER BOOK.

4.1 A potentiometer is a (fixed/variable) resistor.

4.2 An LED changes electrical energy mainly into (light/heat).

4.3 The (anode/cathode) of an LED is marked by a flat edge on its case.

4.4 A stepper motor turns in (steps/continuous turns only).

4.5 A Raspberry Pi GPIO pin can read (digital/analogue) signals directly.

(5)

[5]

TOTAL SECTION A: [35]

SECTION B

QUESTION 5

5.1 Name FIVE main parts of a 3D printer. (5)

5.2 Explain what the infill setting of a 3D print controls. (3)

5.3 Give FOUR safety rules for working with a 3D printer. (4)

[12]

QUESTION 6

A learner uses a Raspberry Pi to control a small conveyor belt model. A green LED shows that the conveyor is running and a red LED shows that it has stopped. A START button and a STOP button control it.

Study the code below and answer the questions that follow.

```
 1  from gpiozero import LED, Button
 2  from time import sleep
 3
 4  green = LED(17)
 5  red = LED(27)
 6  start_button = Button(2)
 7  stop_button = Button(3)
 8
 9  running = False
10  count = 0
11
12  while True:
13      if start_button.is_pressed:
14          running = True
15      if stop_button.is_pressed:
16          running = False
17      if running:
18          green.on()
19          red.off()
20          count = count + 1
21          print("Conveyor running, cycle", count)
22          sleep(1)
23      else:
24          green.off()
25          red.on()
26          sleep(0.1)
```

6.1 Name the library used to control the GPIO pins, and say why it makes the code easier to write. (2)

6.2 What do the statements green = LED(17) and start_button = Button(2) do? (3)

6.3 What does the variable running store, and which lines change its value? (3)

6.4 Describe what the two LEDs do when running is True, and when it is False. (3)

6.5 START is pressed and released, and the program then runs for about 5 seconds. Roughly how many "cycle" messages are printed, and why? (3)

6.6 Why is there a short sleep(0.1) in line 26? (3)

6.7 What happens if both buttons are pressed at the same time? Explain. (3)

6.8 Suggest how a buzzer on GPIO 22 could be added so that it sounds while the conveyor is stopped. Name the gpiozero class you would use. (3)

[23]

QUESTION 7

7.1 Define the term electric current. (2)

7.2 A 12 V battery is connected to a motor with a resistance of 4 Ω. Calculate the current. (3)

7.3 Calculate the power used by the motor in QUESTION 7.2. (3)

7.4 The motor runs for 2 hours. Calculate the energy used, in watt-hours and in kilowatt-hours. (3)

7.5 A 6 Ω resistor and a 3 Ω resistor are connected in parallel. Calculate the total resistance. (3)

7.6 The parallel pair in QUESTION 7.5 is connected to a 6 V supply. Calculate the total current. (2)

7.7 Explain the difference between an open circuit and a short circuit. (4)

[20]

QUESTION 8

Study the scenario below and answer the questions.

> A mining company uses a robot to inspect dangerous tunnels. After a flood, a miner is trapped deep in a tunnel. The supervisor orders the robot to drive into the flooded tunnel to take a rescue line to the miner. The water could damage the robot beyond repair.

8.1 State the Third Law of Robotics. (3)

8.2 Should the robot follow the order, even though it may be destroyed? Justify your answer with reference to all THREE laws of robotics. (7)

[10]

TOTAL SECTION B: [65]

TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

Ohm's law: $V = I \times R$	Power: $P = V \times I = I^2 \times R = V^2 \div R$
Energy: $E = P \times t$	1 kWh = 1 000 Wh
Series: $R_T = R_1 + R_2 + R_3$	Parallel: $1 \div R_T = 1 \div R_1 + 1 \div R_2$; for two resistors $R_T = (R_1 \times R_2) \div (R_1 + R_2)$
Speed: $v = d \div t$	Force: $F = m \times a$; Weight: $W = m \times g$